CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of B. Tech. (ME/EE/CL) Examination

November-December 2013

ME 101.01 Engineering Graphics

Date: 06.12.2013, Friday **Time: 10:00 a.m. To 01:00 a.m.** **Maximum Marks: 70**

Instructions:

1. Attempt all the questions.
2. The question paper comprises of two sections.
3. Section I and II must be attempted in separate answer sheets.
4. Figures to the right indicate the full marks of the question
5. Figures drawn in the question paper are not to the scale.
6. Assume missing data or dimension and mention it clearly.
7. Take suitable scale whenever required and mention it clearly.
8. Use of scientific calculator is allowed.
9. Dimensions shown in the figure are in mm.
10. Show all the necessary dimensions in the solution.

SECTION-I

Q.1 (a) Choose appropriate answer by rewriting the question. **[05]**

1. A line joining any point on spiral with the pole is called as,
   (a) radius vector (b) vectorial angle (c) position vector (d) convolution
2. The object is completely cut by a sectional plane at different positions gives a circular lamina, the object is called,
   (a) cylinder (b) sphere (c) cone (d) pyramid
3. A point whose front view and top view are above XY is situated in,
   (a) first angle (b) second angle (c) third angle (d) fourth angle
4. A point is below HP and in front of VP; the point is in which quadrant.
   (a) 1^{st} (b) 2^{nd} (c) 3^{rd} (d) 4^{th}
5. A pyramid is cut by a plane parallel to its base removing apex, the remaining part is known as
   (a) truncated (b) frustum (c) sectioned (d) prism

Q. 1(b) Answer the following questions. **[10]**

1. Define the representative fraction. **[02]**
2. Construct a scale 2 cm = 1m to read decimeters, to measure maximum distance of 6 m. Show on it a distance of 4 m and 6 dm. **[05]**
3. A point P is 30 mm above H.P. & 30 mm in front of V.P. Draw the projections of point P. **[03]**

OR

Q.1 (b) The major axis and minor axis of an ellipse are 120 mm and 70mm long respectively. Find out the foci and draw half ellipse by rectangular method and other half by concentric circle method. Draw normal and tangent at any point on the curve. **[10]**

Q. 2 A fan is hanging in the center of the room of 4m X 3m X 3m height. The center of the fan is 0.75 m below the ceiling. The switch of this fan is on 3m X 3m size wall at the center height and 0.5 m from the adjacent wall. Find the distance between the fan center and the switch. **[10]**

OR

Q. 2 A line AB, 75mm long, has one end A in VP and other end B is 15 mm above HP. Draw the projections of the line when line is inclined 30° to HP and 60° to VP. **[10]**

Q.3 Fig. 1 shows pictorial view of an object. Use 1st angle projection method and draw (1) Front view in the direction 'X', (2) Top view. **[10]**

SECTION - II

Q.4 A regular hexagonal plate 30 mm side is resting on one of its corners in HP. The diagonal through that corner is inclined at 30° to HP and the plan of that diagonal is inclined to VP by 45°. **[10]**

OR

Q.4 A square ABCD of 50 mm side has its corner A in the H.P., its diagonal AC inclined at 30° to the H.P. and the diagonal BD inclined at 45° to the V.P. and parallel to the H.P. Draw its projections. **[10]**

Q. 5 (a) A cone of base diameter 60 mm and height of 75 mm is resting on HP on its base in such a way that axis is inclined at 30° to HP and (i) the plan of axis is inclined at 45° to VP, (ii) axis is inclined at 45° to VP. Draw projection of the cone when apex is nearer to VP. **[10]**

OR

(a) A cylinder of 50 mm diameter of base and 75 mm height of axis has one of its ends on the H.P. Its cut by and A.I.P. in such a way that the true shape of the section is an ellipse of largest possible major axis. Draw the sectional plane, true shape, development of remaining cut cylinder and also find the inclination of the sectional plane. **[10]**

Q. 5 (b) Give the application of AutoCAD. **[05]**

OR

(b) Difference between Engineering Drawing and Engineering Graphics. **[05]**

Q.6 Fig. 2 shows the orthographic projections of an object in first angle projection system. Draw its isometric view. **[10]**

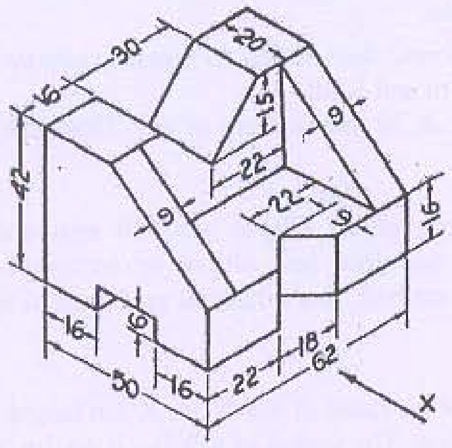

Figure 1

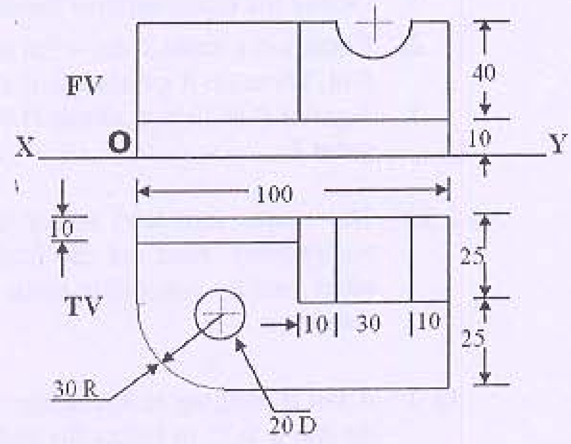

Figure 2